

Optics in the Rope Framework

Classical Interference and Diffraction as Genuine Rope-Wave Superposition — and the Single-Photon Boundary

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Abstract

Optics is one of the rope model's clearest classical successes, and this paper states both the success and its precise limit. Because light in the rope framework is a real transverse wave travelling along physical ropes, classical interference and diffraction arise as genuine MECHANICAL wave superposition: two contributions meet and add or cancel exactly as ripples do. The rope model reproduces the standard two-slit intensity law and the diffraction of light around an obstacle with no additional assumptions — this is the domain where the model is not merely adequate but vivid, because it supplies a literal physical wave where wave optics traditionally posits an abstract field. We then state the boundary plainly: this success is confined to the CLASSICAL, many-photon regime. The single-photon build-up of an interference pattern, one indivisible detection at a time, is NOT reproduced by the rope model, for the reason established elsewhere in the corpus — single-particle interference is the interference of a probability amplitude, which a mechanical wave does not supply. Classical wave optics: reproduced. Single-photon optics: the documented quantum boundary. This paper is careful to keep the two apart.

Scope of this paper

SUCCEEDS: classical interference and diffraction (two-slit intensity law, obstacle diffraction) as real rope-wave superposition — a genuine, vivid in-domain result.

BOUNDARY: single-photon interference (one indivisible detection at a time building a pattern) is NOT reproduced; it requires amplitude interference, the programme's documented quantum limit. This paper claims the classical regime only.

[CARRIER-IDENTITY NOTE (2026-08-05): this paper's classical results (the two-slit law, obstacle diffraction, non-dispersive propagation) are linear-order statements about the superposition of a real transverse wave and stand as written. The MICROSCOPIC identity of that transverse carrier — which mode of the registered two-strand mechanics it is — is an OPEN question per FND-REL-005 and the G/H/I/J carrier campaign, whose finding is that no registered two-strand mode is both transverse-structured and correctly dynamical (the carrier channel-map claim, EM-RECON-022). The classical optics here does not depend on that resolution; it is flagged so paper and registry stay in sync.]

1. Light as a Rope Wave

In the rope framework light is a transverse disturbance propagating along the two-strand ropes that connect matter — a real wave on a real medium, with amplitude, wavelength, and phase. This is the feature that makes classical optics natural in the model: where conventional wave optics works with an abstract electromagnetic field, the rope model works with a literal mechanical wave, and the standard optical phenomena follow from ordinary superposition of such waves. The purpose of this paper is to make that reproduction explicit for the canonical cases, and then to mark exactly where it stops.

2. Classical Interference: The Two-Slit Law

When light from a coherent source reaches two slits, the rope waves emerging from the two openings superpose. Two waves of equal amplitude and wavelength, differing only in the path length from each slit to a point on the screen, combine to give an intensity that varies with the path difference $\Delta = d_1 - d_2$:

$$I(\Delta) \propto 2 I_0 [1 + \cos(2\pi \Delta / \lambda)] \quad (2.1)$$

This is the standard two-slit intensity law, with bright fringes where the path difference is a whole number of wavelengths (waves in phase, crest on crest) and dark fringes where it is a half-integer number (crest on trough, cancellation). The rope model reproduces (2.1) exactly, because it is nothing more than the superposition of two real waves — the same computation as for any classical wave on a medium.

The cancellation here is real and mechanical. At a dark fringe the two rope waves genuinely cancel: crest meets trough, and the medium is momentarily undisplaced there, its energy redistributed to the bright fringes. This is honest physical interference, the kind a mechanical wave model is entitled to, and it is why classical optics is a natural home for the rope picture.

3. Classical Diffraction: Light Around an Obstacle

The same superposition principle reproduces diffraction. Light passing a narrow obstacle or aperture spreads and forms fringes because the rope waves from the unobstructed regions superpose with a continuous range of path differences. The demonstration attributed to the programme's originator — a laser directed past a fine needle, splitting the light into two coherent portions that interfere — is exactly this: a physical wave, split and recombined, producing fringes by ordinary superposition. As a statement about CLASSICAL light, this is correct, and the rope model captures it directly.

4. The Boundary: Single-Photon Interference

The classical success above is confined to the many-photon regime, where there is a real wave with real amplitude to superpose. The boundary appears when the light is attenuated to single photons sent one at a time. Experimentally, an interference pattern still builds up from individual, indivisible detections — and it builds regardless of how long one waits between photons, so it cannot be photons interfering with one another. Each single photon’s landing probability behaves as though it interfered with itself.

Why the rope model cannot reproduce this. A rope wave interferes only when there is amplitude present to superpose; a single indivisible photon provides none to spare, yet the pattern forms. Reproducing it requires the interference of a PROBABILITY AMPLITUDE — a quantity that carries phase and can cancel in the likelihood of a lone particle’s arrival — which a mechanical wave, and more generally the programme’s configuration-counting measurement rule, does not supply. This is the same boundary established in the measurement and non-local-dynamics papers: counting is not interference.

That the single photon is genuinely indivisible — not a weak continuous wave misread by the detector — is independently confirmed by the anticorrelation (g^2) experiment: behind a beam-splitter, a divisible wave would trip both detectors together ($g^2 \geq 1$), whereas single-photon sources measure g^2 far below 1. The rope model’s classical-wave account of light therefore cannot be extended to the single-photon regime by treating the photon as a faint wave; that route is closed by experiment.

5. The Double-Slit Ladder

The success and the boundary are seen together across the versions of the double-slit experiment. The rope model succeeds at the classical version and meets one underlying limitation across the quantum versions:

Version	Physics	Rope model
1. Classical laser light	wave superposition	SUCCEEDS — genuine rope-wave interference, Eq. (2.1)
2. Single photons one at a time	single-particle self-interference	Boundary — needs amplitude interference
3. Which-path detection	measurement / Born rule	Boundary — counting is not interference
4. Delayed choice	no pre-set definite path	Boundary — rope assumes a definite path
5. Quantum eraser	entangled-pair correlation	Boundary — same limit as entanglement

The four boundary cases are one missing ingredient — amplitude interference — in four settings, not four independent failures. The classical case (Version 1) is a full success. This is precisely the programme’s bounded shape: strong on classical wave optics, halting at single-particle quantum optics.

6. Status of Each Result

Result	Status
Two-slit intensity law $I \propto 1 + \cos(2\pi\Delta/\lambda)$	Modeled — exact classical wave superposition
Classical diffraction around an obstacle	Modeled — same superposition principle
Single-photon interference build-up	Boundary — not reproduced (needs amplitude interference)
Photon indivisibility ($g^2 < 1$) as a wave	Failed for the wave picture — closes the weak-wave escape
Which-path / delayed choice / eraser	Boundary — the documented quantum limit

Conclusion

Optics shows the rope programme at its clearest, in both directions. Because light in the model is a real wave on a physical medium, classical interference and diffraction follow as genuine mechanical superposition: the two-slit intensity law and obstacle diffraction are reproduced exactly, with real cancellation at the dark fringes, and the model supplies a literal wave where conventional optics posits an abstract field. That is a vivid in-domain success. The boundary is equally clear: the single-photon regime, where an interference pattern assembles from indivisible detections, is not reproduced, because it requires the interference of a probability amplitude that a mechanical wave — and the programme's counting measurement rule — cannot provide, and because the photon's measured indivisibility closes the escape of treating it as a faint wave. Classical wave optics reproduced; single-photon optics at the documented quantum boundary. Keeping those two apart is the point, and it places optics exactly where the bounded programme says it should be.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **EM-004 [Derived]:** Classical two-slit interference $I \propto 1 + \cos$ [benchmark: benchmarks/optics/optics_checks.py]
- **EM-005 [Derived]:** Non-dispersive Maxwell propagation [benchmark: benchmarks/optics/optics_checks.py]
- **EM-006 [Modeled]:** Birefringence structure [benchmark: benchmarks/optics/optics_checks.py]

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.